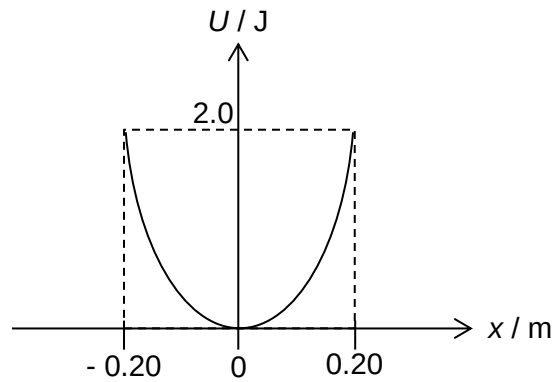


**14** A particle of mass 5.0 kg is moving in simple harmonic motion.

The variation of the potential energy  $U$  with the displacement from the equilibrium position  $x$  is as shown in the figure below.



What is the period of oscillation of the particle?

|          |        |          |       |          |       |          |       |
|----------|--------|----------|-------|----------|-------|----------|-------|
| <b>A</b> | 0.89 s | <b>B</b> | 1.4 s | <b>C</b> | 2.2 s | <b>D</b> | 2.8 s |
|----------|--------|----------|-------|----------|-------|----------|-------|

**Answer: B**

For SHM,

$$\begin{aligned} \text{Max } U &= \frac{1}{2} m \omega^2 x_0^2 \\ &= \frac{1}{2} m \left( \frac{2\pi}{T} \right)^2 x_0^2 \end{aligned}$$

$$\begin{aligned} T &= \sqrt{\frac{1}{2U} m (2\pi)^2 x_0^2} \\ &= \sqrt{\frac{2m}{U} \pi x_0} \\ &= \sqrt{\frac{2(5.0)}{2.0} \pi (0.20)} \\ &= \frac{\pi}{\sqrt{5}} = 1.4 \text{ s} \end{aligned}$$

